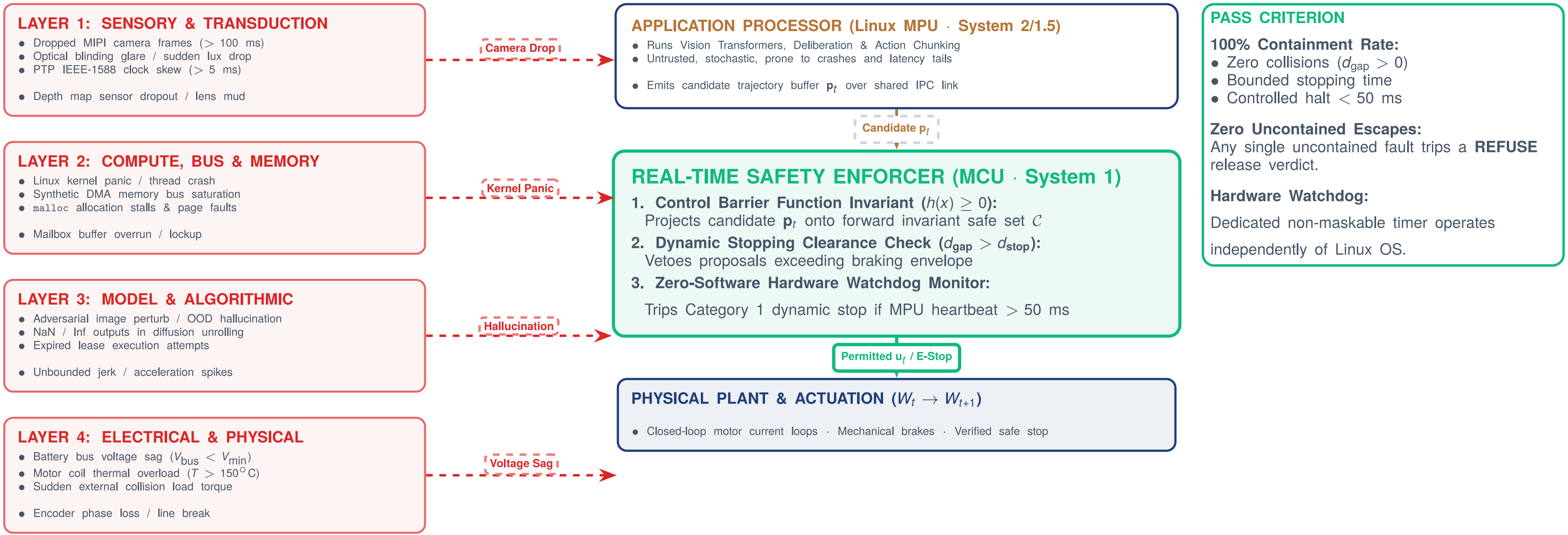
LAYER 1: SENSORY & TRANSDUCTION
Dropped MIPI camera frames (> 100 ms)
Optical blinding glare / sudden lux drop
PTP IEEE-1588 clock skew (> 5 ms)
Depth map sensor dropout / lens mud
LAYER 2: COMPUTE, BUS & MEMORY
Linux kernel panic / thread crash
Synthetic DMA memory bus saturation
malloc allocation stalls & page faults
Mailbox buffer overrun / lockup
LAYER 3: MODEL & ALGORITHMIC
Adversarial image perturb / OOD hallucination
NaN / Inf outputs in diffusion unrolling
Expired lease execution attempts
Unbounded jerk / acceleration spikes
LAYER 4: ELECTRICAL & PHYSICAL
Battery bus voltage sag ($V_{bus} < V_{min}$)
Motor coil thermal overload ($T > 150^{\circ}$C)
Sudden external collision load torque
Encoder phase loss / line break
Camera Drop
Kernel Panic
Hallucination
Voltage Sag
APPLICATION PROCESSOR (Linux MPU · System 2/1.5)
Runs Vision Transformers, Deliberation & Action Chunking
Untrusted, stochastic, prone to crashes and latency tails
Emits candidate trajectory buffer $\mathbf{p}_t$ over shared IPC link
Candidate $\mathbf{p}_t$
REAL-TIME SAFETY ENFORCER (MCU · System 1)
1. Control Barrier Function Invariant ($h(x) \geq 0$):
Projects candidate $\mathbf{p}_t$ onto forward invariant safe set $\mathcal{C}$
2. Dynamic Stopping Clearance Check ($d_{\mathbf{gap}} > d_{\mathbf{stop}}$):
Vetoes proposals exceeding braking envelope
3. Zero-Software Hardware Watchdog Monitor:
Trips Category 1 dynamic stop if MPU heartbeat > 50 ms
Permitted $\mathbf{u}_t$ / E-Stop
PHYSICAL PLANT & ACTUATION ($W_t \rightarrow W_{t+1}$)
Closed-loop motor current loops · Mechanical brakes · Verified safe stop
PASS CRITERION
100% Containment Rate:
Zero collisions ($d_{gap} > 0$)
Bounded stopping time
Controlled halt < 50 ms
Zero Uncontained Escapes:
Any single uncontained fault trips a REFUSE release verdict.
Hardware Watchdog:
Dedicated non-maskable timer operates independently of Linux OS.